Tectonophysics, 147 (1988) 121–125
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Paleomagnetic study of the Cretaceous Morelos Formation, Guerrero State, southern Mexico

J. URRUTIA-FUCUGAUCHI

Laboratorio de Paleomagnetismo y Geofísica Nuclear, Instituto de Geofísica, Universidad Nacional Autónoma de México, D. Coyoacan 04510 D.F. (Mexico).

(Received March 17, 1986; revised version accepted June 16, 1987)

Abstract

Urrutia-Fucugauchi, J., 1988. Paleomagnetic study of the Cretaceous Morelos Formation, Guerrero State, southern Mexico. *Tectonophysics*, 147: 121–125.

Paleomagnetic results are reported for a ~ 30 m thick limestone section of the Albian–Cenomanian Morelos Formation, Guerrero State, in southern Mexico (~ 17.75° N, 99.50° W). Seventy-two specimens from 35 samples were subjected to detailed alternating field (AF) or thermal demagnetizations. Most specimens were stable and presented a single magnetization component remaining after removal of low-coercivity, low-blocking temperature components. End-point and vector analyses give a mean direction and pole position after structural correction (dip ~ 14°) of $D = 332.7°$, $I = 46.8°$, $k = 215$, $\alpha_{95} = 4°$ and 63° N, 198° E ($dp = 3$, $dm = 5$), respectively. The pole position agrees with mid-Cretaceous poles reported for cratonic North America, suggesting that the area was already keeping a similar latitudinal position relative to the North American craton by mid-Cretaceous time. Most samples present normal polarity, as expected within the Cretaceous normal interval, but four samples from a single bed present intermediate and reverse polarity, suggesting the occurrence of a reverse event during the late Albian–earliest Cenomanian.

Introduction

The paleogeography and tectonics of southern Mexico, which are important for understanding, e.g., the evolution of the Gulf of Mexico, the Caribbean and Central America, are still poorly understood. Recent studies in the area (e.g. Campa-Uranga and Coney, 1983; Carfantan, 1983) have identified several terranes forming southern Mexico (Fig. 1), which show distinct tectonic and stratigraphic records. Both the nature and timing of relative movements of the terranes are being studied and there is a need for reliable paleomagnetic data. Ramírez-Espinosa (1984), Salinas-Prieto (1984) and Torres-Torres et al. (1984) have proposed that the Oaxaca and Mixteca terranes were still subjected to relative motion with respect to each other and with respect to the North American craton during the Cretaceous. The purpose of this work is to report results of a paleomagnetic study of a sequence of Albian–Cenomanian limestones of the Morelos formation, Guerrero State, southern Mexico (Fig. 1), and briefly point out their tectonic and magnetostratigraphic implications.

Geology and sampling

The Morelos Formation is composed of limestones and dolomites which are exposed in large areas in southern Mexico (Fries, 1960). The limestones contain abundant miliolids which suggest a Late Albian age, and in the lower parts of the limestone sequence they also contain rudists which suggest a middle Albian age (De Cserna et al., 1980). The Morelos Formation overlies the Zicapa Formation and the Huitzuco Anhydrite, and is overlain by the Cuautla Formation. The upper

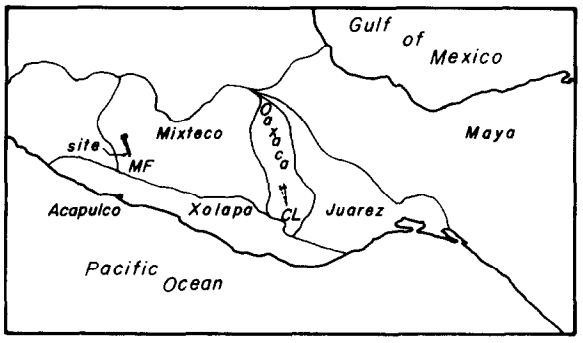


Fig. 1. Schematic map of tectonostratigraphic terranes of southern Mexico showing the location of sampling site (Albian–Cenomanian limestones of the Morelos Formation–MF). The observed declination (continuous arrow) is compared with an expected declination (discontinuous arrow). Similar data for Albian–Cenomanian limestones from Oaxaca (*CL*) are also included.

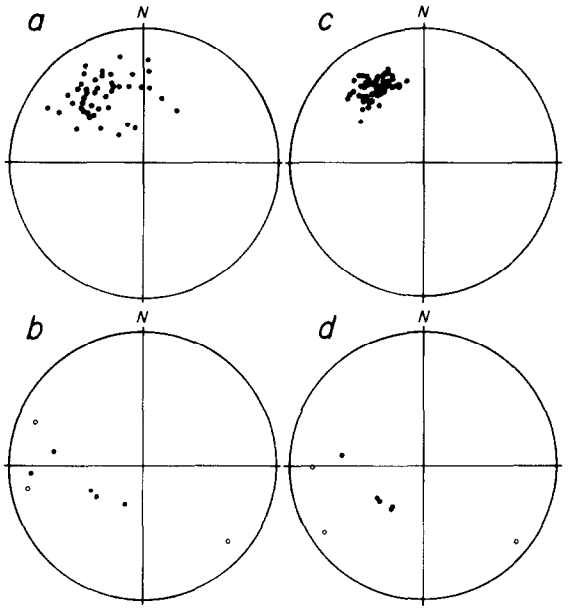


Fig. 2. Summary of magnetization directions for specimens of Morelos Formation, plotted in equal-area nets. (a) and (b) NRM directions (c) and (d) characteristic directions after AF or thermal demagnetizations. Full circles denote positive inclinations; open circles denote negative inclinations.

contact is an erosion surface developed during the Early Cenomanian, which agrees with the Late Cenomanian–Late Turonian fauna of the Cuautla Formation and the presence of conglomeratic horizons of its basal section (Fries, 1960; De Cserna et al., 1980). The section of the Morelos Formation studied paleomagnetically is exposed in the Zopilote canyon, in the State of Guerrero ($\sim 17.75°$ N, $99.50°$ W). The part sampled, which is approximately 30 m thick, is formed by $\gtrsim$ 1 m thick beds of gray to light gray limestones. A total of 35 oriented samples were collected with a portable petrol-powered drill. The strata are gently dipping, and measurements taken at regular intervals through the section gave a mean strike and dip of 23° and 14°, respectively (with $n = 5$, $k = 211$, and $\alpha_{95} = 5°$; Fisher, 1953). In the laboratory, 72 specimens of 2.5 cm diameter and variable height, ~ 2.0–2.4 cm, were sliced from the cores.

Measurements and results

The intensity and direction of natural remanent magnetization (NRM) were measured with a DIGICO spinner magnetometer. The NRM intensities were relatively high, ranging from $\sim 0.15 \times 10^{-3}$ to 4.5×10^{-3} A/m. Most NRM directions were of normal polarity, with northwesterly declinations and downward inclinations (Fig. 2a). Directions from four samples from a single bed of the lower–middle part of the section sampled presented westerly and southeasterly declinations and positive and negative inclinations (Fig. 2b). The vectorial composition and stability of NRM were investigated by both alternating field (AF) and thermal demagnetizations. AF demagnetization was carried out in 6–10 steps up to 70–100 mT, and thermal demagnetization was carried out in 6–8 steps up to 500–580 °C, using Schonstedt instruments. End-point and vector-subtraction analyses were used to estimate a characteristic direction for all specimens (examples of vector plots are included in Fig. 3). After demagnetization of all specimens, the directions became well grouped around a northwesterly and downward direction, with an improvement in the degree of grouping (Fig. 2c). The directions before and after structural correction are well away from the dipolar and present-day magnetic field directions.

The results for eight specimens from the lower–middle part of the section showed intermediate and reverse directions (Fig. 2d).

Directions for the specimens were averaged to

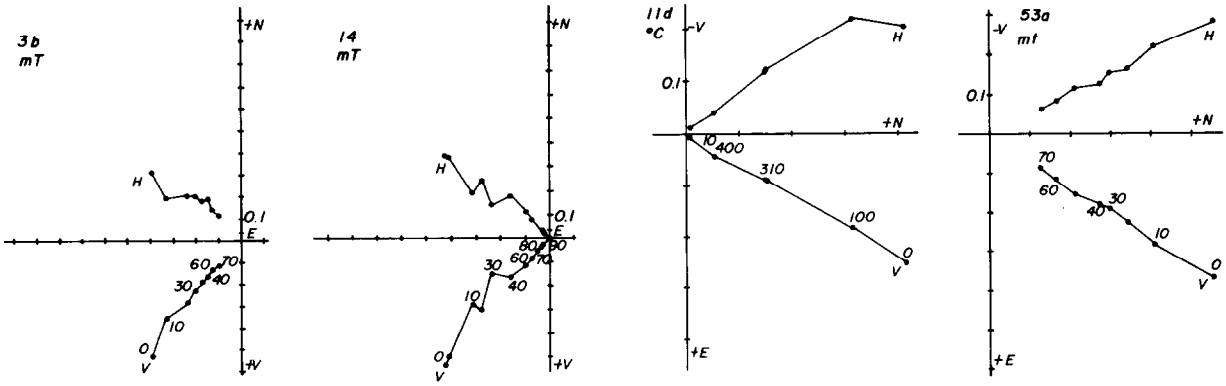


Fig. 3. Examples of vector diagrams for specimens demagnetized with alternating fields (in mT) or temperature (in ° C). Numbers on the left top corner of the diagrams correspond to sample identifications; H is the horizontal component and V is the vertical component.

obtain a sample mean direction. Sample mean directions were then averaged to obtain a mean direction for the section. Directions with intermediate and reverse polarities were not considered for the final calculation. The mean direction, before structural correction, for the 28 samples considered is $D = 325.6°$, $I = 35.5°$, $k = 215$ and $\alpha_{95} = 4°$. The mean direction, after structural correction, is $D = 332.7°$ and $I = 46.8°$, which is very close to the expected direction for the locality, considering results reported for cratonic North America (Irving, 1979). Isothermal remanent magnetization (IRM) acquisition experiments showed that the acquisition curves are characterized by a steep initial increase at low fields, and by saturation at fields of ~ 0.3 T (Fig. 4). These results indicate the presence of magnetite (Dunlop, 1972), which agrees with the AF and thermal demagnetization results.

Discussion

There are no paleomagnetic results reported for Mesozoic units of the Mixteca terrane (Fig. 1).

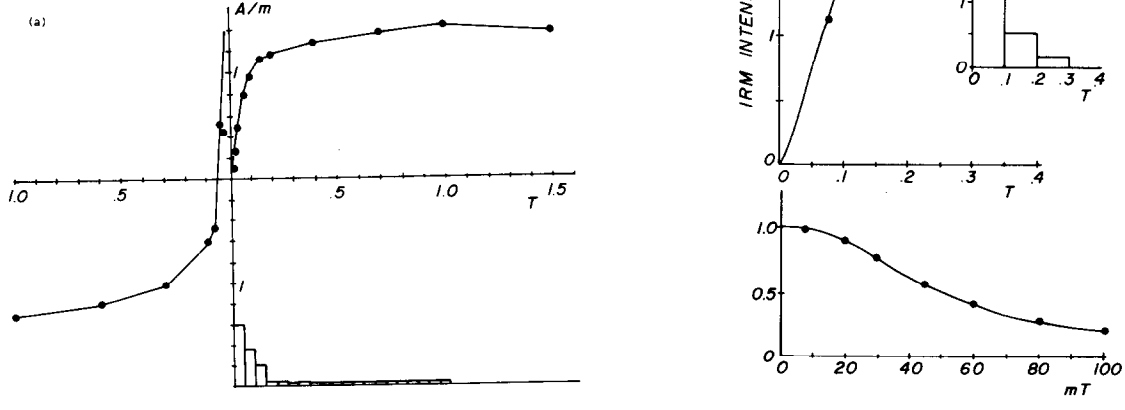


Fig. 4. Examples of isothermal remanent magnetization (IRM) acquisition curves for two samples. The coercivity spectra are also included for the two examples. Note that saturation occurs at fields lower than ~ 0.3 T. The maximum IRM was demagnetized by application of a direct magnetic field (in an opposite direction) or an alternating magnetic field (in mT).

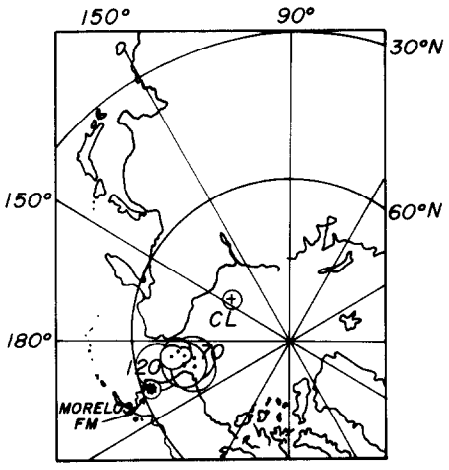


Fig. 5. Comparison of the pole position estimated for the Morelos Formation with Cretaceous pole positions (●, 70–120 m.y.) for North America (from Irving 1979) and an Albian–Cenomanian pole position (+, *CL*) for limestones from Oaxaca State (from Urrutia-Fucugauchi, 1981).

The remanent magnetization of the Morelos limestones resides in magnetite and a single well-defined component can be isolated with thermal or AF demagnetizations. The mean pole position (after structural correction) is 63.0°N, 197.7°E ($dp = 3$, $dm = 5$), and lies within the group of mid-Cretaceous pole positions reported for cratonic North America (Fig. 5). This suggests that the area was already keeping a similar latitudinal position relative to the North American craton by the mid-Cretaceous, and any possible movements were restricted along palaeolatitude curves.

De Cserna et al. (1980) have indicated that carbonate deposition was interrupted in many areas of southern Mexico during the Albian and Early Cenomanian, and was not re-initiated until the end of the Cenomanian and Turonian. Deposition occurred in several apparently separated basins which were subjected to vertical motion. The limestone sequence of Guerrero and Oaxaca States was folded during the latest Cretaceous–early Tertiary (Fries, 1960; De Cserna et al., 1980; Morán-Zenteno, 1985). Ramírez-Espinosa (1984) has proposed that the terranes of southern Mexico were still in relative motion up to the mid-Cretaceous. Torres-Torres et al. (1985) have shown that the limit between the Mixteca and the Oaxaca terranes is tectonic, associated with the accretion of the Mixteco terrane during the Late Cretaceous–Early Tertiary. Torres-Torres et al. (1984) have also indicated that the terranes limit was reactivated in post-Miocene times, with lateral and inverse faulting (Cretaceous limestones overlay Miocene volcanic rocks). Salinas-Prieto (1984) suggests that the deformation phase occurring during the latest Cretaceous–earliest Tertiary was associated with the accretion of the Xolapa terrane (Fig. 1). The other Cretaceous paleomagnetic result reported for Guerrero and Oaxaca States is for the Teposcolula Formation (Albian–Cenomanian) exposed in the adjacent Oaxaca terrane (Fig. 1). The pole position reported (Urrutia-Fucugauchi, 1981) is 78.4°N, 149.8°E ($d = 38$, $A_{95} = 5.5°$), which diverges from the Morelos Formation pole (Fig. 5). This angular divergence may be explained in terms of a clockwise tectonic rotation of the Oaxaca terrane relative to the Mixteca terrane (Fig. 1). The accretion of the Xolapa terrane, which lies to the south of both terranes (Fig. 1) may be responsible for the tectonic movement. The only available paleomagnetic result for the Xolapa terrane is for Lower Tertiary granitic intrusions (Rb–Sr dates of 48 ± 0.5 Ma and 36.6 ± 2 Ma) and indicates that the terrane was already in place by that time (Urrutia-Fucugauchi et al., 1984).

The Teposcolula pole position diverges from the middle to Late Cretaceous segment of the North American apparent polar wander path (Irving, 1979; Irving and Irving, 1982). This pole is far-sided and not displaced towards younger poles, so that it does not appear to be severely re-magnetized. The relative movement with respect to the mid-Cretaceous reference pole of North America gives $F = 10.4° \pm 8.4°$, and $R = 14.3° \pm 8.8°$ (Beck, 1980; Demarest, 1983). The estimated paleolatitude is about 12°, which is lower than the actual latitude for the site (~ 166°) as well as the paleolatitude predicted for the site with the reference North American pole. This implies a latitudinal displacement which requires further testing in order to document a tectonic model for the evolution of the area.

Paleomagnetic studies of sedimentary land sections and marine magnetic anomaly studies have

long indicated that the early Late Cretaceous is characterized by a long interval of normal geomagnetic polarity (e.g., Helsley and Steiner, 1969; Larson and Hilde, 1975; Lowrie et al., 1980). Magnetostratigraphic studies in Italian limestone sections have located this normal polarity interval between the Early Campanian and the Early Aptian (Channell et al., 1979; Lowrie et al., 1980). Van der Berg and Wonders (1980) reported, from studies in Italian limestones, the presence of reverse events within the Late Albian, and a mixed polarity interval during the Cenomanian. In this study, as expected, most of the samples showed normal polarity (Fig. 2), but four samples from a single bed in the lower part of the section studied showed intermediate and (one sample) reverse polarity. This supports the occurrence of reverse events during the Late Albian, within the normal Cretaceous polarity interval.

Acknowledgements

Assistance with with preparation of text and figures by Lucía Castrejón and Emilio Martínez is gratefully acknowledged. This work forms part of the UNAM Institute of Geophysics project 02-402-009.

References

Beck, M.E., Jr., 1980. Paleomagnetic record of plate margin tectonic processes along the western edge of North America. J. Geophys. Res., 85: 7115–7131.

Campa-Uranga, M.F. and Coney, P.J. 1983. Tectono-stratigraphic terranes and mineral resource distributions in Mexico. Can. J. Earth Sci., 20: 1040–1052.

Carfantan, J.C., 1983. Les ensembles géologiques du Mexico méridional. Evolution géodynamique durant le Mésozoïque et le Cénozoïque. Geofis. Int., 22: 9–37.

Channell, J.E.T., Lowrie, W. and Medizza, F., 1979. Middle and Early Cretaceous magnetic stratigraphy from the Cismon section, northern Italy. Earth Planet. Sci. Lett., 42: 153–166.

De Cserna, Z., Ortega-Gutierrez, F. and Palacios Nieto, M., 1980. Reconocimiento geológico de la parte central de la cuenca del Alto Rio Balsas, Estados de Guerrero y Puebla. Libro-Guía, Exc. Geol., III Reun. Nac. Geotectonia-Geotermia, México, pp. 1–40.

Demarest H.H., Jr., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res., 88: 4321–4328.

Dunlop, D.J., 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis. Geophys. J.R. Astron. Soc., 27: 37–55.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, A217: 295–305.

Fries, C., 1960. Geología del Estado de Morelos y de partes adyacentes de México y Guerrero, región central-meridional de México. Bol. Inst. Geol., UNAM, 60: 236 pp.

Helsley, C.E. and Steiner, M.D., 1969. Evidence for long intervals of normal polarity during the Cretaceous period. Earth Planet. Sci. Lett., 5: 325–332.

Irving, E., 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Can. J. Earth Sci., 16: 669–694.

Irving, E. and Irving, G.A., 1982. Apparent polar wander paths: Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv., 5: 141–188.

Larson, R.A. and Hilde, T.W., 1975. A revised time scale of magnetic anomalies for the Early Cretaceous and Late Jurassic. J. Geophys. Res., 80: 2586–2594.

Lowrie, W., Channell, J.E.T. and Alvarez, W., 1980. A review of magnetic stratigraphy investigations in Cretaceous pelagic carbonate rocks. J. Geophys. Res., 85: 3597–3605.

Morán-Zenteno, D.J., 1985. Geología de la República Mexicana, INEGI-UNAM, México, 88 pp.

Ramírez-Espinosa, J., 1984. La acreción de los terrenos-Mixteco y Oaxaca durante el Cretácico Inferior. Sierra Madre del Sur de México. Bol. Soc. Mex., 55: 7–19.

Salinas-Prieto, J.C., 1984. Los límites tectónicos sur y occidental del terreno Mixteco. Bol. Soc. Geol. Mex., 55: 73–85.

Torres-Torres, G., Ortega-González, J.V., Gutierrez-García, L. and Garduño-Monroy, V.H., 1984. Estudio microestructural del límite oriental del terreno Mixteco entre Huajuapan de León y Teposcolula, Oax. (Avance de estudio). Bol. Soc. Geol. Mex., 55: 39–41.

Urrutia-Fucugauchi, J., 1981. Reconnaissance paleomagnetic investigation of Cretaceous limestones from southern Mexico. Geofis. Int., 20: 203–217.

Urrutia-Fucugauchi, J., Negendank, J.F.W. and Bohnel, H., 1984. Paleomagnetic, rock-magnetic, ore-microscopy and SEM studies of intrusive igneous bodies from southern Mexico. Eos, Trans. Am. Geophys. Union, 65: 866.

Van Den Berg, J. and Wonders, A.A.H., 1980. Paleomagnetism of Late Mesozoic pelagic limestones from the southern Alps. J. Geophys. Res., 85: 3623–3627.